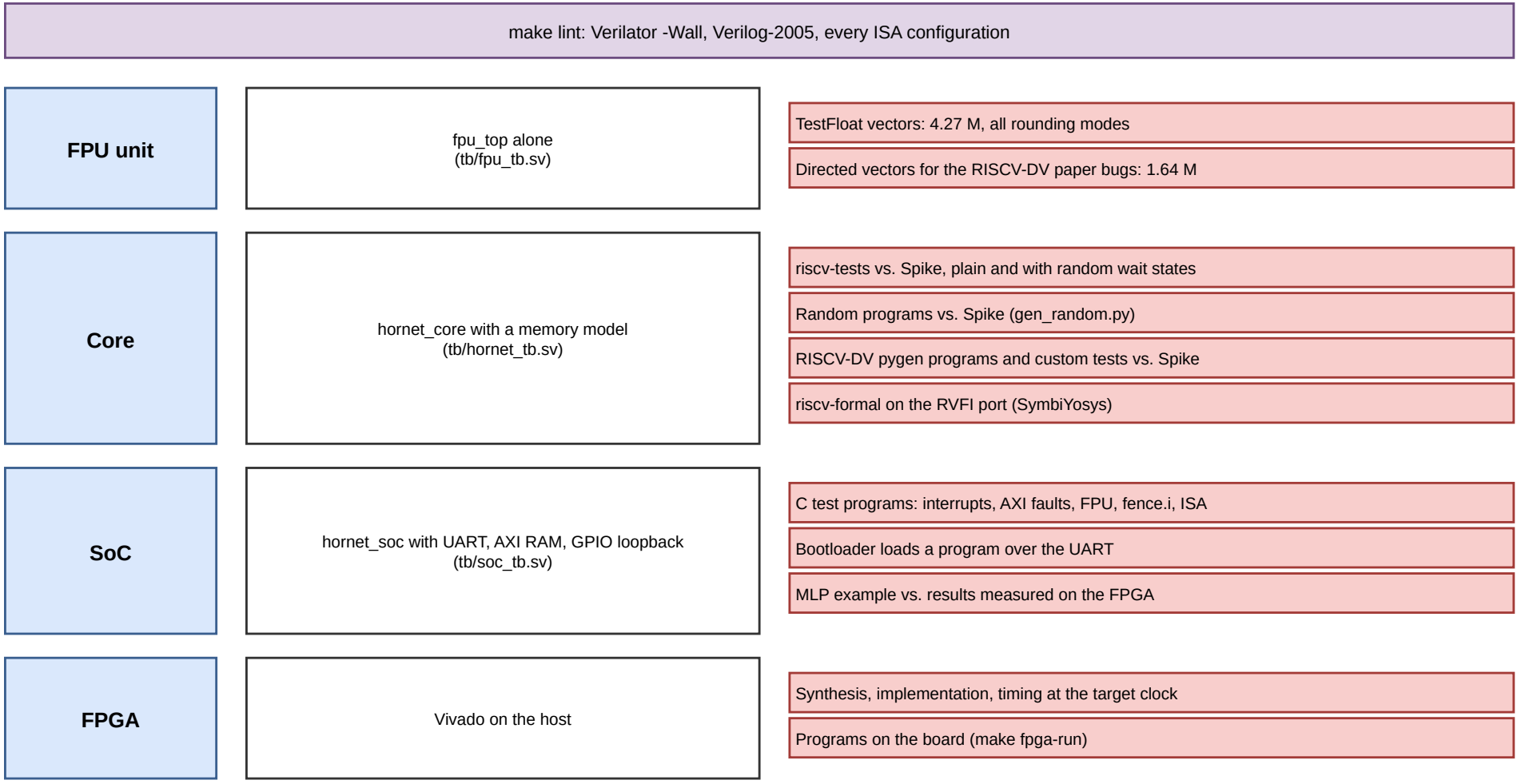
make lint: Verilator -Wall, Verilog-2005, every ISA configuration
FPU unit
fpu_top alone
(tb/fpu_tb.sv)
TestFloat vectors: 4.27 M, all rounding modes
Directed vectors for the RISCV-DV paper bugs: 1.64 M
Core
hornet_core with a memory model
(tb/hornet_tb.sv)
riscv-tests vs. Spike, plain and with random wait states
Random programs vs. Spike (gen_random.py)
RISCV-DV pygen programs and custom tests vs. Spike
riscv-formal on the RVFI port (SymbiYosys)
SoC
hornet_soc with UART, AXI RAM, GPIO loopback
(tb/soc_tb.sv)
C test programs: interrupts, AXI faults, FPU, fence.i, ISA
Bootloader loads a program over the UART
MLP example vs. results measured on the FPGA
FPGA
Vivado on the host
Synthesis, implementation, timing at the target clock
Programs on the board (make fpga-run)